

Contrôle de calcul 24 mars.

1. $(-15) - (-13) = (-15) + (+13) = \underline{-2}$

2. $[(5-9) + (3-5)] - [(7+3-6) - (7-10)] = [-4 + -2] - [5 - (-3)]$
 $= -6 - [8]$
 $= \underline{-14}$

3. $(-10) + (-15) - (-30) + (+7) - (+12)$
 $= (-10) + (-15) + (+30) + (+7) + (-12)$
 $= (30+7) - (10+15+12)$
 $= 37 - 37$
 $= \underline{0}$

4. $-\frac{1}{2} + \frac{3}{4} - \frac{5}{8} = -\frac{4}{8} + \frac{6}{8} - \frac{5}{8} = \frac{-4+6-5}{8} = \underline{-\frac{3}{8}}$

5. $(+3) \times (+7) \times (-5) \times (-4) = + (3 \times 7 \times 5 \times 4) = +21 \times 20 = \underline{+20}$

6. $\frac{5}{9} \times \frac{-7}{5} = \frac{5 \times (-7)}{9 \times 5} = \underline{-\frac{7}{9}}$

7. $(-56) \div (-8) = \underline{+7}$

8. $\frac{-63}{70} = \frac{-7 \times 9}{7 \times 10} = \underline{-\frac{9}{10}}$

9. $\frac{-3 + \frac{3}{4} - \frac{1}{3}}{5 + \frac{2}{5} - \frac{2}{3}} = \frac{\frac{-3 \times 12}{12} + \frac{3 \times 3}{4 \times 3} - \frac{1 \times 4}{3 \times 4}}{\frac{5 \times 15}{15} + \frac{2 \times 3}{5 \times 3} - \frac{2 \times 5}{3 \times 5}} = \frac{\frac{-36+9-4}{12}}{\frac{75+6-10}{15}} = \frac{-\frac{31}{12}}{\frac{71}{15}} = -\frac{31}{12} \times \frac{15}{71}$
 $= \underline{-\frac{31 \times 15}{12 \times 71}} = \underline{-\frac{31 \times 3 \times 5}{3 \times 4 \times 71}} = \underline{-\frac{31 \times 5}{4 \times 71}} = \underline{-\frac{155}{284}}$

10. $\frac{1}{2} + \frac{3}{2} \times \frac{3}{4} = \frac{1}{2} + \frac{9}{8} = \frac{4}{8} + \frac{9}{8} = \underline{\frac{13}{8}}$

$$11. (+3)^4 = 3 \times 3 \times 3 \times 3 = 9 \times 9 = \underline{81}$$

$$12. (-2)^3 = (-2) \times (-2) \times (-2) = \underline{-8}$$

$$13. \left(-\frac{2}{3}\right)^2 = \frac{-2}{3} \times \frac{-2}{3} = \frac{-2 \times -2}{3 \times 3} = \underline{\frac{4}{9}}$$

$$14. xy \times x^2 \times ay = \underline{ax^3y^2}$$

$$15. \left(-\frac{3}{5}a^2\right)\left(\frac{2}{3}b^2x\right)(-x^4) = \left(-\frac{3}{5} \times \frac{2}{3} \times -1\right)a^2b^2x^5 = \underline{\frac{2}{5}a^2b^2x^5}$$

$$\begin{aligned} 16. (3x-5) + [2x-5-(3x-2y+4)-(4x-3y-9)] \\ = 3x-5 + [2x-5-3x+2y-4-4x+3y+9] \\ = 3x-5 + [(2-3-4)x + (2+3)y -5-4+9] \\ = 3x-5 + (-5)x + 5y + 0 \\ = \underline{-2x+5y-5} \end{aligned}$$

$$17. \left(-\frac{2}{5}ab^3\right)^2 = \left(-\frac{2}{5}ab^3\right) \times \left(-\frac{2}{5}ab^3\right) = \frac{-2 \times -2}{5 \times 5} a^2b^3b^3 = \underline{\frac{4}{25}a^2b^6}$$

$$\begin{aligned} 18. (2x-3y)(4x-2) &= 2x \times 4x + 2x \times (-2) + (-3y) \times 4x + (-3y) \times (-2) \\ &= \underline{8x^2 - 4x - 12xy + 6y} \end{aligned}$$

$$19. (x+1)x - 2(x+1) = (x+1)x + (x+1)(-2) = (x+1)(x-2)$$

$$20. 4x^2 + 5x + yx = x \times 4x + x \times 5 + x \times y = \underline{x(4x+5+y)}$$

Contrôle de calcul du 24 mars

1. $(-17) - (-13) = -17 + 13 = \underline{-4}$.

2. $[(7-9) + (3-5)] - [(7+3-5) - (7-10)] - [-2 + -2] - [5 - (-3)]$
 $= -4 - 8 = \underline{-12}$.

3. $(-11) + (-15) - (-30) + (+7) - (+12) = (-11) + (-15) + (+30) + (+7) + (-12)$
 $= -37 - 38$
 $= \underline{-1}$.

4. $-\frac{1}{2} + \frac{3}{4} - \frac{5}{8} = \frac{-4}{8} + \frac{6}{8} - \frac{5}{8} = \frac{-4+6-5}{8} = \underline{-\frac{3}{8}}$.

5. $(+3) \times (+7) \times (-4) \times (-4) = (21) \times (+16) = \underline{336}$.

6. $\frac{5}{8} \times \frac{-7}{5} = \frac{5 \times -7}{8 \times 5} = \underline{-\frac{7}{8}}$.

7. $(-56) \div (-4) = \underline{+14}$.

8. $-\frac{63}{70} = \frac{-7 \times 9}{7 \times 10} = \underline{-\frac{9}{10}}$.

9. $-\frac{3 + \frac{3}{4} - \frac{1}{3}}{5 + \frac{2}{5} - \frac{2}{3}} = \frac{\frac{-3 \times 12}{12} + \frac{3 \times 3}{4 \times 3} - \frac{1 \times 4}{3 \times 4}}{\frac{5 \times 15}{15} + \frac{2 \times 3}{5 \times 3} - \frac{2 \times 5}{3 \times 5}} = \frac{\frac{-36 + 9 - 4}{12}}{\frac{75 + 6 - 10}{15}} = \frac{\frac{-31}{12}}{\frac{71}{15}} = -\frac{31 \times 15}{12 \times 71} = -\frac{31 \times 5 \times 3}{3 \times 4 \times 71}$
 $= -\frac{31 \times 5}{4 \times 71} = \underline{-\frac{155}{284}}$.

10. $\frac{1}{2} + \frac{3}{2} \times \frac{2}{5} = \frac{1}{2} + \frac{3}{5} = \frac{5}{10} + \frac{6}{10} = \frac{11}{10}$. (-7/5)

11. $(+2)^5 = 2 \times 2 \times 2 \times 2 \times 2 = 4 \times 4 \times 2 = 16 \times 2 = \underline{32}$.

12. $(-3)^3 = (-3) \times (-3) \times (-3) = +9 \times (-3) = \underline{-27}$.

$$13. \left(\frac{-3}{4}\right)^2 = \frac{-3}{4} \times \frac{-3}{4} = \underline{\frac{+9}{16}}$$

$$14. xy \times y^2 \times ax = \underline{ax^2y^3}$$

$$15. \left(\frac{-3}{5}a^2\right)\left(\frac{2}{3}b^2x\right)(-x^4) = \left(\frac{-3}{5} \times \frac{2}{3} \times -1\right) a^2b^2x^5 = \underline{+\frac{2}{5}a^2b^2x^5}$$

$$\begin{aligned} 16. (3x-5) + [3x-6 - (3x-2y+4) - (4x-3y-8)] \\ = 3x-5 + [3x-6 - 3x+2y+4 - 4x+3y+8] \\ = 3x-5 + [3x-3x-4x+2y+3y-6-4+8] \\ = 3x-5 + [-4x+5y-1] \\ = 3x-4x+5y-5-1 \\ = \underline{-x+5y-6} \end{aligned}$$

$$17. \left(\frac{-3}{5}ab^3\right)^2 = \frac{-3}{5}ab^3 \times \frac{-3}{5}ab^3 = \left(\frac{-3}{5} \times \frac{-3}{5}\right) aab^3b^3 = \underline{\frac{9}{25}a^2b^6}$$

$$\begin{aligned} 18. (2x-3y)(5x-2) &= 2x \times 5x + 2x \times (-2) + (-3y) \times 5x + (-3y) \times (-2) \\ &= \underline{10x^2 - 4x - 15xy + 6y} \end{aligned}$$

$$19. (x+3)x - 2(x+3) = (x+3)x + (-2)(x+3) = (x+3)(x-2)$$

$$20. 3x^2 + yx + 2x = x(3x + y + 2)$$